

AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements



EXAMPLE CANDIDATE RESPONSE OF EXTERNAL SUMMATIVE ASSESSMENT

12 grade

CHEMISTRY
Paper 3

Astana, 2025 year

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Introduction

This booklet presents a collection of example candidate responses from the External Summative Assessment in Chemistry for Grade 12 students at Nazarbayev Intellectual Schools. The assessment is based on the content and expectations of the international Advanced Level Chemistry curriculum.

The aim of this booklet is to support students and teachers in preparing for future assessments by providing real student responses, examiner comments, and mark allocations. The examples are taken from Papers 2 and 3 and are categorized into three levels of performance: A, C, and E, as described in the Chemistry Test Specification.

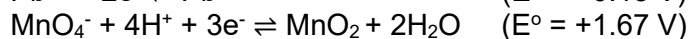
Examiners marked student work using the official mark scheme, which includes standard abbreviations such as *ecf* (error carried forward). This is applied when a student makes an error in an earlier part of their response, but subsequent steps remain logically valid. The number in square brackets indicates the maximum mark for each response.

By reviewing the sample answers and examiner feedback in this booklet, users can gain a clearer understanding of the assessment criteria, identify common strengths and errors, and enhance teaching approaches in line with the NIS–Programme — the Educational Programme of Nazarbayev Intellectual Schools AEO for Grades 11 and 12 — and international standards.

Question 1 (a), (b)

An experiment is set up to investigate how concentration affects the electrode potential of an electrochemical cell.

The first experiment is performed under standard conditions. The half-equations of the electrochemical cell are shown.



(a) Write the balanced ionic equation for the overall cell reaction.

.....

.....

.....

..... [1]

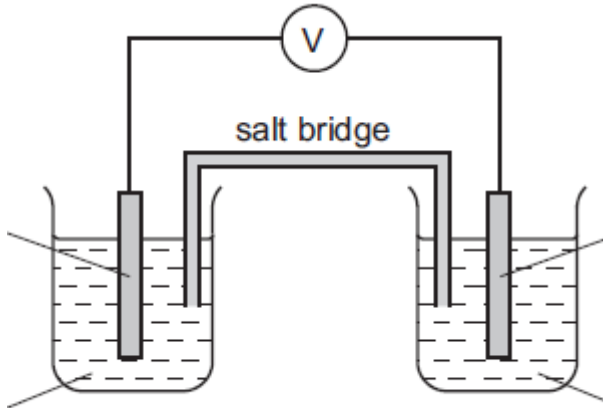
(b) (i) Draw a diagram of the apparatus you would use to measure the standard electrode potential for the lead/manganate (VII) ion cell. You should label all the electrodes and solutions.

[3]

(ii) Calculate the standard cell potential (E_{cell}) of this cell.

$E_{\text{cell}} = \dots\dots\dots \text{ V} \quad [1]$

Mark Scheme

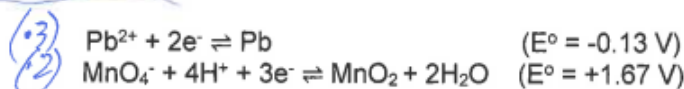
Question	Answer	Marks	Additional Guidance
1(a)	$3\text{Pb} + 2\text{MnO}_4^- + 8\text{H}^+ = 3\text{Pb}^{2+} + 2\text{MnO}_2 + 4\text{H}_2\text{O}$	[1]	Must be balanced
1(b)(i)	 M1 two beakers connected by salt bridge and Voltmeter M2 for correct half-cell $\text{Pb}/\text{Pb}^{2+}(\text{aq})$ M3 for correct half-cell $\text{MnO}_4^-(\text{aq}), \text{H}^+(\text{aq})/\text{MnO}_2$ and Pt	1 1 1 [3]	M3 is not accepted without H^+ ion Accept two correct half cells without (aq) – 1 mark
1(b)(ii)	+ 1.80V	[1]	Accept 1.8

Example candidate response—grade A

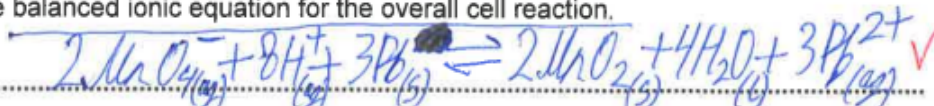
- 1 An experiment is set up to investigate how concentration affects the electrode potential of an electrochemical cell.

For
Examiner
Use

The first experiment is performed under standard conditions. The half-equations of the electrochemical cell are shown.

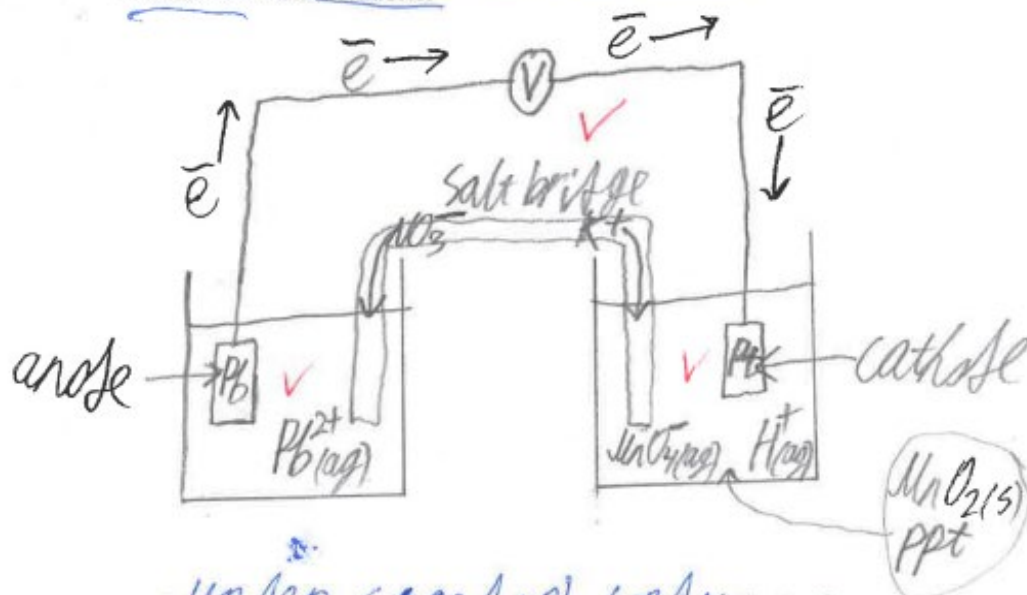


- (a) Write the balanced ionic equation for the overall cell reaction.



[1]

- (b) (i) Draw a diagram of the apparatus you would use to measure the standard electrode potential for the lead/manganate(VII) ion cell. You should label all the electrodes and solutions.



[3]

- (ii) Calculate the standard cell potential (E_{cell}) of this cell.

$$1.67 - (-0.13) = 1.8 \text{ V}$$

$$E_{\text{cell}} = 1.8 \text{ V} \quad [1]$$

Examiner's comment

The candidate correctly selects oxidation and reduction half-equations, combines them into an overall equation, and balances it correctly. For this, they receive 1 mark.

By drawing a diagram of the apparatus, the candidate includes all necessary details, such as the voltmeter, salt bridge, labelled electrodes, and all aqueous and solid substances involved in the process. As a result, they receive the maximum of 3 marks for this task.

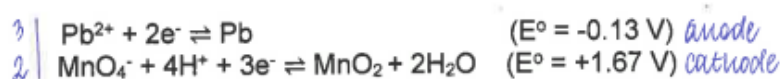
In addition, the candidate correctly calculates the standard cell potential and is awarded 1 mark.

Example candidate response—grade C

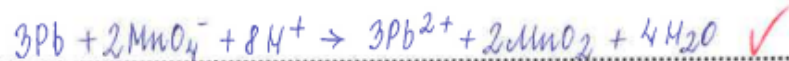
An experiment is set up to investigate how concentration affects the electrode potential of an electrochemical cell.

For
Examin
Use

The first experiment is performed under standard conditions. The half-equations of the electrochemical cell are shown.

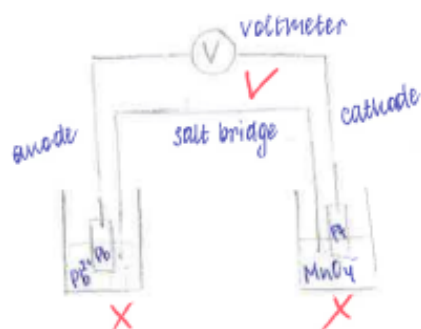


(a) Write the balanced ionic equation for the overall cell reaction.



[1]

(b) (i) Draw a diagram of the apparatus you would use to measure the standard electrode potential for the lead/manganate(VII) ion cell. You should label all the electrodes and solutions.



[3]

(ii) Calculate the standard cell potential (E_{cell}) of this cell.

$$E_{\text{cell}}^{\circ} = 1.67 - (-0.13) = 1.80 \text{ V}$$

$$E_{\text{cell}} = 1.80 \text{ V} \quad \checkmark \quad [1]$$

Examiner's comment

The candidate correctly selects oxidation and reduction half- equations, combines them into an overall equation, and balances it correctly. For this, they receive 1 mark.

By attempting to draw a diagram of the apparatus, the candidate correctly constructs the setup, including a salt bridge, a voltmeter connected to electrodes, and two beakers with electrolytes. The candidate selects correct electrodes and shows some ions in the electrolytes. The mistake is that they omit the states of substances in the electrolytes, as well as $\text{H}^+(\text{aq})$ and $\text{MnO}_2(\text{s})$. For this reason, they lose 2 marks and receive only 1 out of 3 marks for this task.

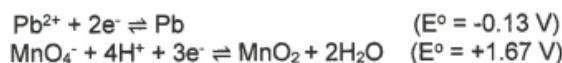
The candidate is awarded 1 mark for correctly calculating the standard cell potential.

Example candidate response—grade E

An experiment is set up to investigate how concentration affects the electrode potential of an electrochemical cell.

For
Examiner's
Use

The first experiment is performed under standard conditions. The half-equations of the electrochemical cell are shown.



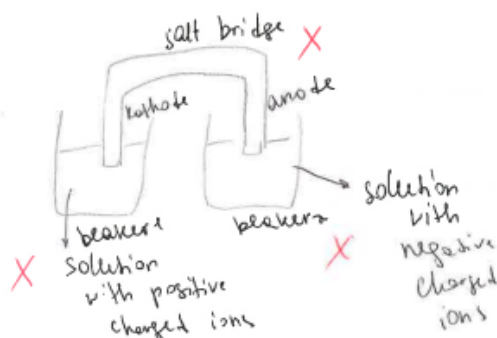
(a) Write the balanced ionic equation for the overall cell reaction.



[1]

0

- (b) (i) Draw a diagram of the apparatus you would use to measure the standard electrode potential for the lead/manganate(VII) ion cell. You should label all the electrodes and solutions.



[3]

0

- (ii) Calculate the standard cell potential (E_{cell}) of this cell.

$$E_{\text{cell}} = 1,67 - (-0,13) = 1,80$$

$$E_{\text{cell}} = 1,80 \text{ V} \quad [1]$$

1

Examiner's comment

The candidate combines two half-equations but does not consider their directions and the number of electrons, and therefore loses 1 mark. By attempting to draw the diagram for measuring the standard cell potential, the candidate includes a salt bridge but fails to connect the beakers with electrodes or link them using a voltmeter.

The candidate draws two beakers with electrolytes but does not show all necessary ions in the electrolytes. Additionally, they don't show the form of the electrodes. As a result, no marks are awarded for the diagram.

The candidate earns 1 mark for correctly calculating the cell potential.

Question 1 (c), (d)(i)

- (c) State the colour change during the experiment.

..... [1]

- (d) The cell potential is measured for various concentrations of MnO_4^- (aq) and the results obtained are shown in the table below.

- (i) Complete the third column of the table. Give each answer **two** decimal places.

concentration of MnO_4^- (aq)/mol dm ⁻³	cell potential / V	log[MnO_4^- (aq)]
---	--------------------	-----------------------------

5.00×10^{-1}	1.794	
1.00×10^{-1}	1.780	
4.00×10^{-2}	1.773	
1.00×10^{-2}	1.761	
5.00×10^{-3}	1.755	
2.00×10^{-3}	1.747	
8.00×10^{-4}	1.739	

[2]

Mark Scheme

Question	Answer	Marks	Additional Guidance																								
1(c)	From pink/purple (solution) to brown (ppt)	[1]	Accept colourless with brown																								
1(d)(i)	<table><tr><th>concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$</th><th>cell potential / V</th><th>Log $[\text{MnO}_4^-(\text{aq})]$</th></tr><tr><td>$5.00 \times 10^{-1}$</td><td>1.794</td><td>-0.30</td></tr><tr><td>1.00×10^{-1}</td><td>1.780</td><td>-1.00</td></tr><tr><td>4.00×10^{-2}</td><td>1.773</td><td>-1.40</td></tr><tr><td>1.00×10^{-2}</td><td>1.761</td><td>-2.00</td></tr><tr><td>5.00×10^{-3}</td><td>1.755</td><td>-2.30</td></tr><tr><td>2.00×10^{-3}</td><td>1.747</td><td>-2.70</td></tr><tr><td>8.00×10^{-4}</td><td>1.739</td><td>-3.10</td></tr></table> Correctly calculated values All data to 2 decimal places	concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential / V	Log $[\text{MnO}_4^-(\text{aq})]$	5.00×10^{-1}	1.794	-0.30	1.00×10^{-1}	1.780	-1.00	4.00×10^{-2}	1.773	-1.40	1.00×10^{-2}	1.761	-2.00	5.00×10^{-3}	1.755	-2.30	2.00×10^{-3}	1.747	-2.70	8.00×10^{-4}	1.739	-3.10	1 1 [2]	
concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential / V	Log $[\text{MnO}_4^-(\text{aq})]$																									
5.00×10^{-1}	1.794	-0.30																									
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1.00×10^{-2}	1.761	-2.00																									
5.00×10^{-3}	1.755	-2.30																									
2.00×10^{-3}	1.747	-2.70																									
8.00×10^{-4}	1.739	-3.10																									

Example candidate response—grade A

- (c) State the colour change during the experiment.

From purple (MnO_4^- solution) to brown (MnO_2 [1] crystals)

- (d) The cell potential is measured for various concentrations of MnO_4^- (aq) and the results obtained are shown in the table below.

- (i) Complete the third column of the table. Give each answer **two** decimal places.

concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential / V	$\log[\text{MnO}_4^-(\text{aq})]$
5.00×10^{-1}	1.794	-0,30
1.00×10^{-1}	1.780	-1,00
4.00×10^{-2}	1.773	-1,40
1.00×10^{-2}	1.761	-2,00
5.00×10^{-3}	1.755	-2,30
2.00×10^{-3}	1.747	-2,70
8.00×10^{-4}	1.739	-3,10

[2]

For
Examiner's
Use

1/1

2/2

Examiner's comment

The candidate correctly states the colour change from purple to brown, receiving 1 mark for this.

They complete the table with correct calculations and give the answer to the required number of decimal places. For this, they receive the maximum mark.

Example candidate response—grade C

- (c) State the colour change during the experiment.

..... *from pink to colourless* [1]

- (d) The cell potential is measured for various concentrations of MnO_4^- (aq) and the results obtained are shown in the table below.

- (i) Complete the third column of the table. Give each answer **two** decimal places.

concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential / V	$\log[\text{MnO}_4^-(\text{aq})]$
5.00×10^{-1}	1.794	<i>-0.30</i>
1.00×10^{-1}	1.780	<i>-1.00</i>
4.00×10^{-2}	1.773	<i>-1.40</i>
1.00×10^{-2}	1.761	<i>-2.00</i>
5.00×10^{-3}	1.755	<i>-2.30</i>
2.00×10^{-3}	1.747	<i>-2.70</i>
8.00×10^{-4}	1.739	<i>-3.10</i>

✓✓ [2]

For
Examin
Use

2

Examiner's comment

The candidate shows colour changes typical of redox reactions in an acidic environment but does not consider the substances given in the task. For this, they lose 1 mark.

The candidate performs correct calculations and gives the answer to the required number of decimal places. For this, they are awarded 2 marks.

Example candidate response—grade E

- (c) State the colour change during the experiment.

~~From purple to~~ rapidly turning brown X [1]

- (d) The cell potential is measured for various concentrations of MnO_4^- (aq) and the results obtained are shown in the table below.

- (i) Complete the third column of the table. Give each answer **two** decimal places.

concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential /V	$\log[\text{MnO}_4^-(\text{aq})]$
5.00×10^{-1}	1.794	-0,30
1.00×10^{-1}	1.780	-1
4.00×10^{-2}	1.773	-1,40
1.00×10^{-2}	1.761	-2
5.00×10^{-3}	1.755	-2,30
2.00×10^{-3}	1.747	-2,70
8.00×10^{-4}	1.739	-3,10

X V [2]

For
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0

1

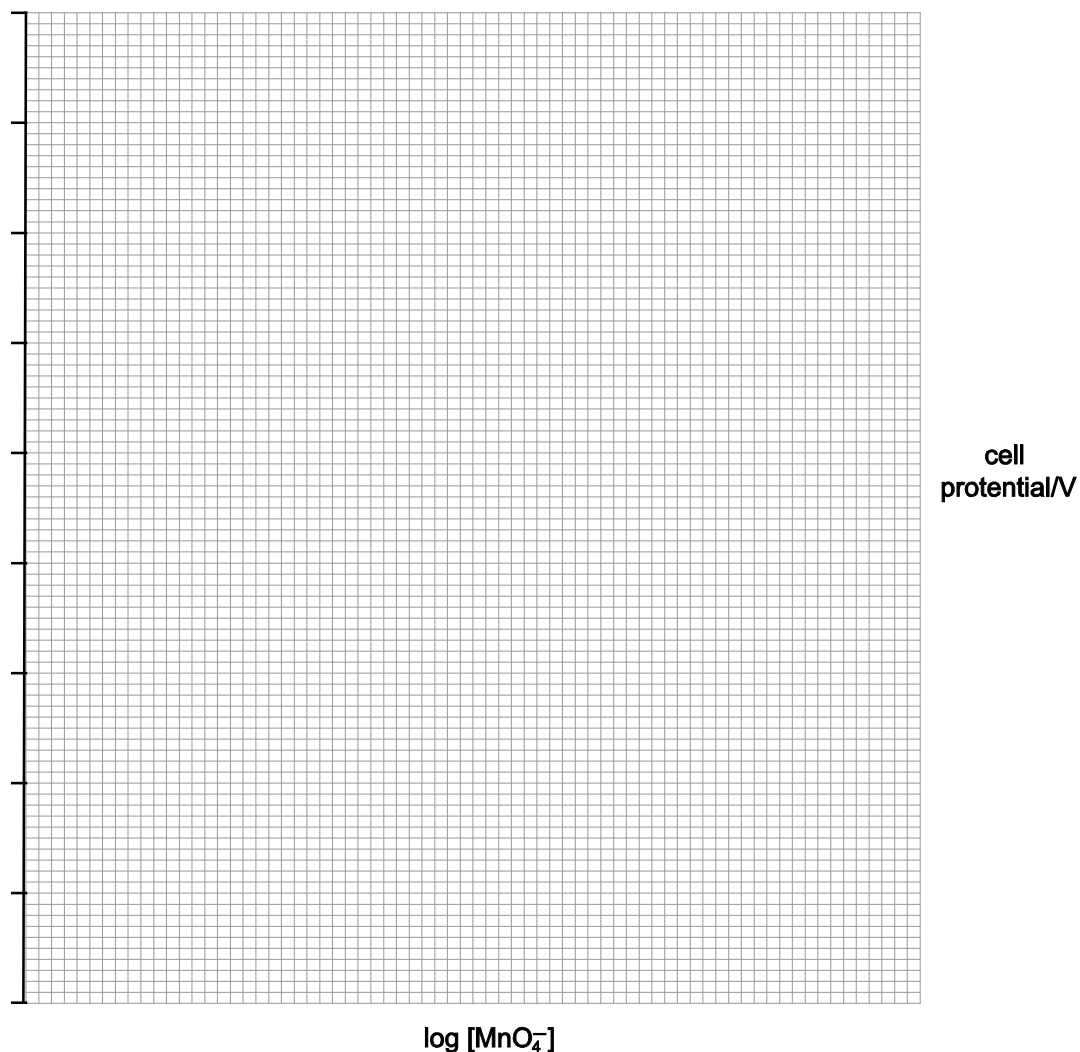
Examiner's comment

The candidate correctly states the brown colour of the product but does not show the appearance of the reagents. For this they lose 1 mark.

By completing the table, the candidate performs the necessary calculations but does not round some values to two decimal places. Therefore, 1 out of 2 marks was awarded.

Question 1 (d)(ii),(iii)

- (ii) Plot a graph to show the relationship between $\log [\text{MnO}_4^- (\text{aq})]$ and the cell potential measured **and** draw the line of best fit.



[2]

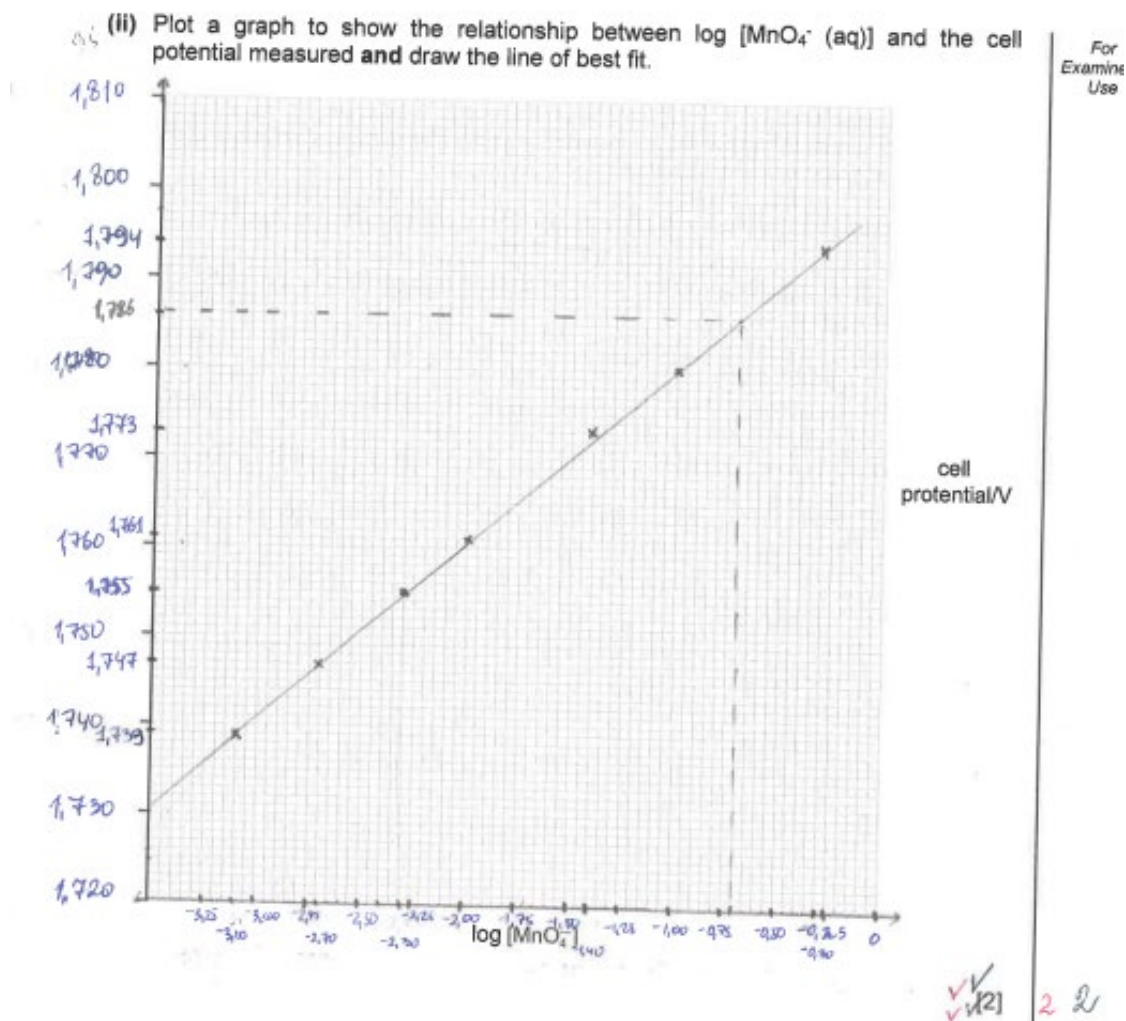
- (iii) Use your graph to determine the concentration of $\text{MnO}_4^- (\text{aq})$ if the cell potential is 1.786 V.

concentration of $\text{MnO}_4^- (\text{aq})$ mol dm⁻³ [1]

Mark Scheme

Question	Answer	Marks	Additional Guidance
1(d)(ii)	Linear scales used that ensure that when plotted the points occupy at least half of the available grid	1	Ecf 1d(i) calculation Log $[\text{MnO}_4^- (aq)]$
	All points correctly and best fit line	1 [2]	
1(d)(iii)	Range 0.150-0.250 OR Any correct calculated value according to students' graph	[1]	

Example candidate response—grade A



- (iii) Use your graph to determine the concentration of $\text{MnO}_4^- (\text{aq})$ if the cell potential is 1.786 V.

$$\log [\text{MnO}_4^-] = -0.70$$

$$[\text{MnO}_4^-] = 10^{-0.70} = 0.1995 \approx 0.20 \text{ M} = 2.00 \cdot 10^{-1}$$

concentration of $\text{MnO}_4^- (\text{aq})$ $2.00 \cdot 10^{-1} (0.20) \text{ mol dm}^{-3}$ ✓[1]

1 1

Examiner's comment

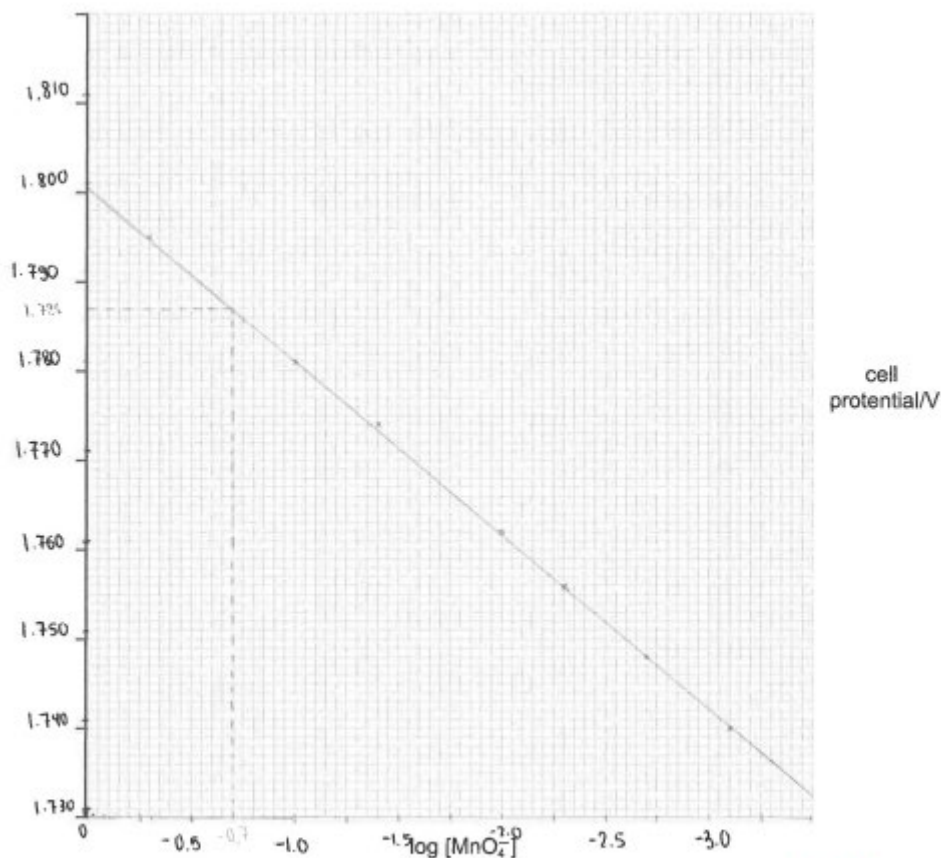
The candidate correctly and neatly draws a scale, placing lower numbers on the left side. The graph covers most of the grid area. For this, they receive 1 mark.

The candidate's data points correspond to the scale. The candidate draws the line of best fit correctly, receiving 1 mark for the this.

The candidate clearly shows their working on the graph when differentiating the logarithm with respect to the given cell potential. They perform calculations and provide the correct answer, receiving 1 mark.

Example candidate response—grade C

- (ii) Plot a graph to show the relationship between $\log [\text{MnO}_4^- (\text{aq})]$ and the cell potential measured and draw the line of best fit.



XV
XV [2]

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1
1

- (iii) Use your graph to determine the concentration of $\text{MnO}_4^-(\text{aq})$ if the cell potential is 1.786 V.

$$E_{\text{cell}} = 1.786 \quad \log [\text{MnO}_4^-] = -0.70$$

$$[\text{MnO}_4^-] = 10^{-0.70} = 0.1995 \text{ mol dm}^{-3}$$

$$2.00 \cdot 10^{-1}$$

concentration of $\text{MnO}_4^-(\text{aq})$ $2.00 \cdot 10^{-1}$ ✓✓ mol dm⁻³ [1]

1/1

Examiner's comment

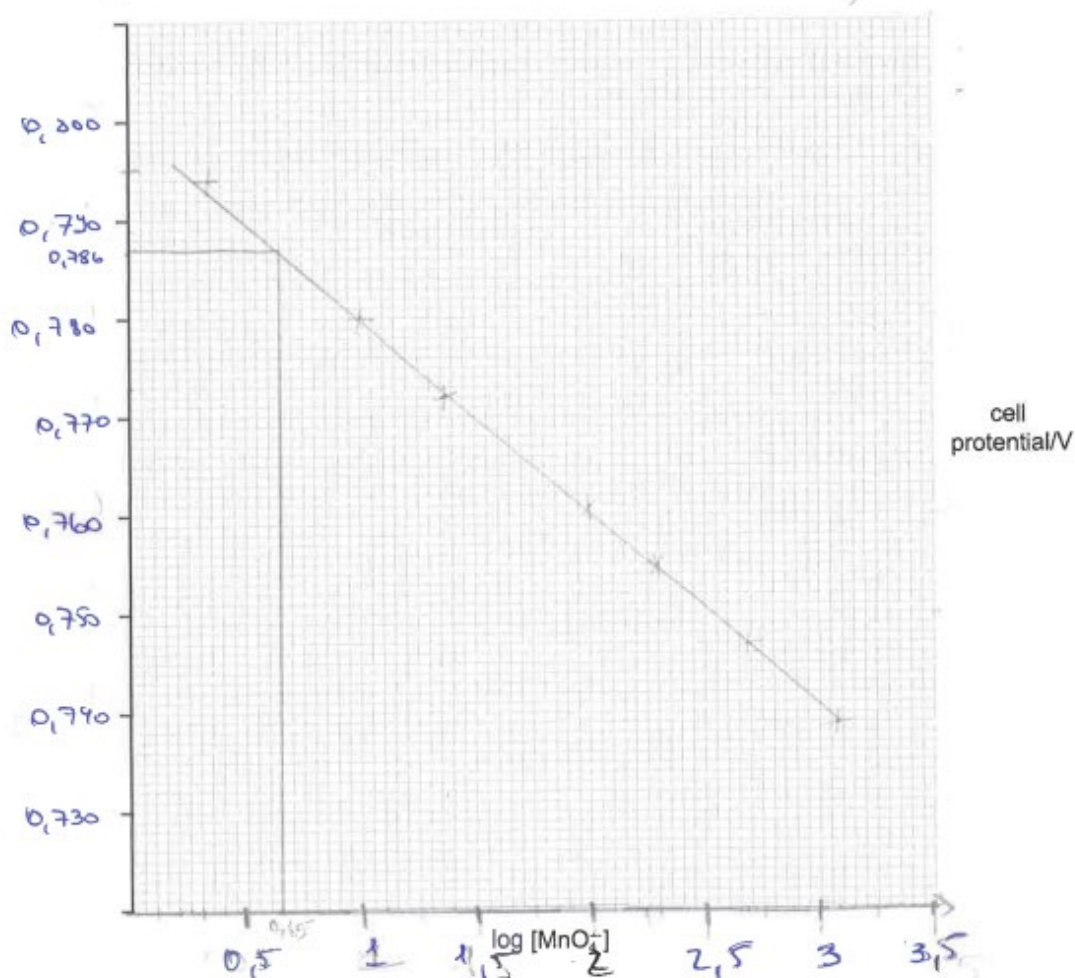
The candidate draws the graph's scale incorrectly, placing the lower values on the right side of the scale. The data points correspond to the scale. The candidate draws the line of best fit correctly, receiving 1 mark for the graph.

The candidate clearly shows their working on the graph when differentiating the logarithm with respect to the given cell potential. They perform calculations and give the correct answer, receiving 1 mark.

Example candidate response—grade E

- (ii) Plot a graph to show the relationship between $\log [\text{MnO}_4^- (\text{aq})]$ and the cell potential measured and draw the line of best fit.

For
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[2]

- (iii) Use your graph to determine the concentration of $\text{MnO}_4^- (\text{aq})$ if the cell potential is 1.786 V.

$$\log (\text{MnO}_4^-) = 0.65$$

$$\sqrt[10]{0.65} = 0.96$$

concentration of $\text{MnO}_4^- (\text{aq})$ $9 \cdot 10^{-1}$ mol dm^{-3} [1]

Examiner's comment

The scale and the plotted points on the candidate's graph do not correspond to the data from the table. Additionally, the value of one of the points does not correspond to the candidate's scale by one grade. As a result, they receive no marks for the graph drawing.

The candidate uses the graph to determine the $\log [\text{MnO}_4^-]$ but has difficulties calculating the concentration from this value. Therefore, no marks are awarded for this task.

Question 1 (e), (f), (g)

(e) State and explain how a decrease in concentration of Pb^{2+} affects the overall cell potential value.

.....

.....

.....

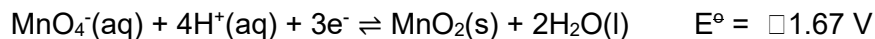
..... [2]

(f) The solutions contained in the two half-cells must be connected using a salt bridge. Explain why potassium chloride solution is not suitable to use as the salt bridge.

.....

..... [1]

(g) A student conducts a second experiment using an electrochemical cell with the two half-equations shown.



The student observes no change in colour during the experiment.

Explain why $\text{Pb}^{2+}(\text{aq})$ will **not** reduce $\text{MnO}_4^{-}(\text{aq})$ to $\text{MnO}_2(\text{s})$.

.....

.....

.....

..... [2]

Mark Scheme

Question	Answer	Marks	Additional Guidance
1(e)	Higher than Ecell value / 1.80	1	
	If concentration of Pb^{2+} decrease equilibrium shift to the right.	1	
		[2]	
1(f)	KCl will form ppt with Pb^{2+}	[1]	
1(g)	Pb^{4+} better oxidizing agent/ Pb^{2+} weaker reducing agent/ higher electrode potential value	1	ORA
	Ecell negative/ reaction not feasible	1	OWTTE
		[2]	

Example candidate response—grade A

- (e) State and explain how a decrease in concentration of Pb^{2+} affects the overall cell potential value.

If $[\text{Pb}^{2+}]$ is decreased, according to Le Chatelier's principle, equilibrium will shift to the right. That will cause ~~more~~ electrons to be emitted more strongly as their conc. will also increase. ~~E_{cell} will for Pb^{1+}/Pb will be~~ more negative, so overall E_{cell} potential value will increase. [2] ✓

- (f) The solutions contained in the two half-cells must be connected using a salt bridge. ✓

Explain why potassium chloride solution is not suitable to use as the salt bridge.

Potassium chloride will react with the solutions. Lead(II) chloride is insoluble. ✓ ✓ [1] ✓

For
Examiner's
Use

2.2

1.1

- (g) A student conducts a second experiment using an electrochemical cell with the two half-equations shown.



The student observes no change in colour during the experiment.

Explain why $\text{Pb}^{2+}(\text{aq})$ will not reduce $\text{MnO}_4^{-}(\text{aq})$ to $\text{MnO}_2(\text{s})$.

Because, now, electrode potential for $\text{Pb}^{4+}/\text{Pb}^{2+}$ half-cell is greater (more positive) than for manganese. This means that current (electrons) will flow from manganese half-cell to lead half-cell as the latter one has greater ability to attract electrons. Pb^{4+} is going to be, thus, reduced and MnO_4^{-} will be oxidised to MnO_4^{-} . Backward reaction (reducing MnO_4^{-} to MnO_2) is not spontaneous as an E_{cell} for it will be negative. ✓✓

[2]

[Total: 16]

22

~~15~~ 14

Examiner's comment

The candidate correctly describes the effect of decreasing the lead ion concentration and explains it by a shift in equilibrium to the right.

The candidate correctly explains why potassium chloride cannot be used in the salt bridge, linking this to formation of insoluble lead(II) chloride.

The candidate compares the electrode potentials of two half-reactions and draws the correct conclusion about the negative value of the cell potential. For this, they receive the maximum mark.

Example candidate response—grade C

- (e) State and explain how a decrease in concentration of Pb^{2+} affects the overall cell potential value.

For
Examin
Use

Decrease in concentration of Pb^{2+} will increase self potential value, because it has opposite to manage relationship with overall self potential. [2]

1

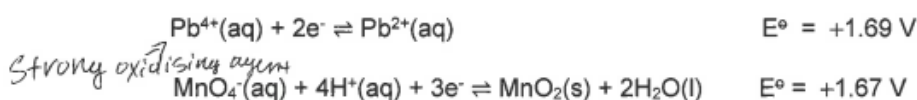
- (f) The solutions contained in the two half-cells must be connected using a salt bridge.

Explain why potassium chloride solution is not suitable to use as the salt bridge.

because lead(II) will form precipitate with chloride ions. $\text{PbCl}_2 \downarrow$ [1]

1

- (g) A student conducts a second experiment using an electrochemical cell with the two half-equations shown.



The student observes no change in colour during the experiment.

Explain why $\text{Pb}^{2+}(\text{aq})$ will **not** reduce $\text{MnO}_4^-(\text{aq})$ to $\text{MnO}_2(\text{s})$.

because its reaction has higher electrode potential, therefore it will & Pb^{2+} will accept electrons poorly. Pb^{2+} is a weak oxidising agent. Therefore, Pb^{4+} will accept electrons and MnO_2 will oxidise to MnO_4^- . [2]

1

[Total: 16]

8

Examiner's comment

The candidate correctly shows the effect of decreasing the lead ion concentration on the overall cell potential but struggles to explain the reason. Therefore, only 1 out of 2 marks is awarded for this part.

The candidate correctly explains why potassium chloride cannot be used in the salt bridge, linking it to the precipitation of lead(II) chloride. As a result, they receive 1 mark.

In describing the impossibility of the reaction, the candidate compares the electrode potentials but does not develop their idea further, receiving only 1 out of 2 marks.

Example candidate response—grade E

- (e) State and explain how a decrease in concentration of Pb^{2+} affects the overall cell potential value.

For
Examin
Use

I + will decrease the overall cell potential value X, because this will decrease the number of electrons

X [2]

- (f) The solutions contained in the two half-cells must be connected using a salt bridge.

Explain why potassium chloride solution is not suitable to use as the salt bridge.

because salt bridge has to be solid X [1]

- (g) A student conducts a second experiment using an electrochemical cell with the two half-equations shown.



The student observes no change in colour during the experiment.

Explain why $\text{Pb}^{2+}(\text{aq})$ will **not** reduce $\text{MnO}_4^{-}(\text{aq})$ to $\text{MnO}_2(\text{s})$.

Because it has a higher E° cell value which means that it is more likely to gain electrons and be reduced itself, X [2]

Examiner's comment

In the question about the effect of lead ion concentration on the cell potential, the candidate does not correctly show the effect and fails to link their explanation to a shift in equilibrium. As a result, no marks were awarded for this part.

In explaining why potassium chloride cannot be used in the salt bridge, the candidate mentions the formation of a solid but fails to identify which component is involved or specify the precipitated substance. Therefore, they receive no marks for this.

In attempting to explain the impossibility of the redox reaction, the candidate uses incorrect terminology and describes a different (priority) reaction, rather than one given in the task. They receive no marks for this part.

Question 2 (a)

A student is given four test-tubes containing colourless organic liquids labelled as **K**, **L**, **M** and **Q**.

The liquids are:

- propanoic acid
- propanone
- propanal
- propan-2-ol.

The student uses experiments to identify the organic liquids.

(a) In the first experiment pieces of magnesium ribbon are separately added to each test-tube, only **K** and **L** produce bubbles.

The second experiment is repeated with magnesium carbonate, only **K** produces bubbles.

(i) Identify compound **K**

..... [1]

(ii) Write a balanced equation of reaction of **K** with magnesium carbonate.

..... [1]

(iii) Give positive test for gas produced with magnesium carbonate.

..... [1]

(iv) Draw the structure of an isomer of **K** which does **not** release gas with carbonate ion.

[1]

Mark Scheme

Question	Answer	Marks	Additional Guidance
2(a)(i)	Propanoic acid/ $\text{CH}_3\text{-CH}_2\text{-COOH}$	[1]	
2(a)(ii)	$2\text{CH}_3\text{CH}_2\text{COOH} + \text{MgCO}_3 \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Mg} + \text{CO}_2 + \text{H}_2\text{O}$	[1]	
2(a)(iii)	Lime water turns milky/ppt	[1]	OWTTE
2(a)(iv)	$\text{HCOOCH}_2\text{CH}_3$ / $\text{CH}_3\text{COOCH}_3$	[1]	Accept $\text{CH}_3\text{CO-CH}_2\text{OH}$ / $\text{CH}_3\text{-CH(OH)-CHO}$ / $\text{HO-CH=CH-CH}_2\text{-OH}$

Example candidate response—grade A

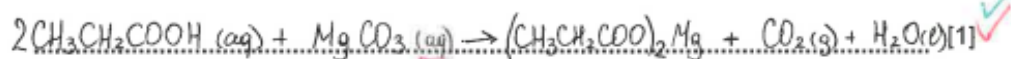
- (a) In the first experiment pieces of magnesium ribbon are separately added to each test-tube, only **K** and **L** produce bubbles.
The second experiment is repeated with magnesium carbonate, only **K** produces bubbles.



- (i) Identify compound **K**

propanoic acid ✓ ✓ [1] 15 1

- (ii) Write a balanced equation of reaction of **K** with magnesium carbonate.

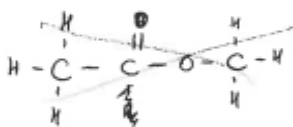


- (iii) Give positive test for gas produced with magnesium carbonate.

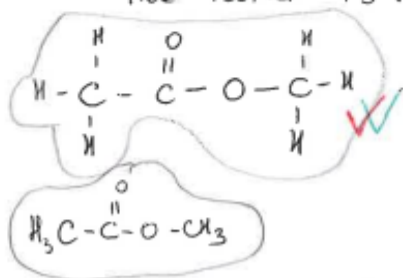
CO₂: passing through lime water ⇒ gives white ppt ✓ ✓ [1] 15 1

- (iv) Draw the structure of an isomer of **K** which does **not** release gas with carbonate ion.

propanoic acid — $\text{CH}_3\text{CH}_2\text{COOH}$ —



The isomer is:



methyl ethanoate ✓ ✓

[1]

15 1

Examiner's comment

The candidate correctly identifies compound **K**, and is awarded 1 mark.

The candidate writes a correctly balanced equation for the reaction of **K** with magnesium carbonate, receiving 1 mark. They also identify the correct positive test for the gas produced, receiving 1 mark.

The candidate accurately draws the structure of an isomer of **K** that does not react with the carbonate ion, for which they are awarded 1 mark.

Example candidate response—grade C

- (a) In the first experiment pieces of magnesium ribbon are separately added to each test-tube, only K and L produce bubbles.
The second experiment is repeated with magnesium carbonate, only K produces bubbles.

(i) Identify compound K

propanoic acid ✓ [1] 1 1

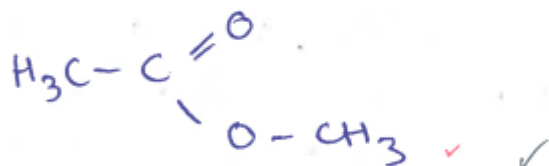
(ii) Write a balanced equation of reaction of K with magnesium carbonate.

$2\text{CH}_3\text{CH}_2\text{COOH} + \text{MgCO}_3 \Rightarrow 2\text{CH}_3\text{CH}_2\text{COO}^- + \text{CO}_2 + 2\text{H}^+$ ✗ [1] 0 0

(iii) Give positive test for gas produced with magnesium carbonate.

white ppt with limewater ✓ [1] 1 1

(iv) Draw the structure of an isomer of K which does **not** release gas with carbonate ion.



[1] 1 1

Examiner's comment

The candidate correctly identifies compound K and received 1 mark.

The candidate provides an incorrect equation for the reaction of K with magnesium carbonate, resulting in the loss of 1 mark. Despite this, they correctly identify the positive test for the gas produced and are awarded 1 mark for this.

The candidate accurately draws the structure of an isomer of K that does not react with the carbonate ion, receiving 1 mark.

Example candidate response—grade E

- (a) In the first experiment pieces of magnesium ribbon are separately added to each test-tube, only K and L produce bubbles.
The second experiment is repeated with magnesium carbonate, only K produces bubbles.

(i) Identify compound K

propanoic acid ✓ [1] 1

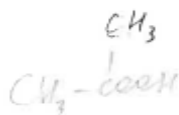
(ii) Write a balanced equation of reaction of K with magnesium carbonate.

$\text{CH}_3\text{CH}_2\text{COOH} + \text{MgCO}_3 \rightarrow \text{CH}_3\text{CH}_2\text{COO}^- + \text{MgCO}_2 + \text{O}_2 + \text{H}^+$ [1] 0

(iii) Give positive test for gas produced with magnesium carbonate.

gives a white ppt with limewater, ppt dissolves with excess CO_2 [1] 1

(iv) Draw the structure of an isomer of K which does **not** release gas with carbonate ion.



X

0

[1]

Examiner's comment

The candidate correctly identifies compound K and receives 1 mark.

The equation for the reaction of K with magnesium carbonate is incorrect, resulting in a loss of 1 mark. However, they correctly state the positive test for the gas produced, receiving 1 mark.

The candidate draws an incorrect isomer of K that does not react with the carbonate ion. As a result, they lose 1 mark for this part.

Question 2 (b)

(b) It is known that **L** and **M** can be oxidised

(i) Suggest a suitable oxidising agent and give conditions.

Reagent [1]

Conditions [2]

(ii) Describe the observation of the reaction between **L** and oxidising in **b(i)**.

.....
..... [1]

(iii) Use the information from (a) and (b) to identify **L**.

compound **L** [1]

Mark Scheme

Question	Answer	Marks	Additional Guidance
2(b)(i)	Oxidizing agent: $K_2Cr_2O_7$ / $KMnO_4$ Conditions: H^+ / H_2SO_4 heat/warming/T	1	Accept names of compounds
		1	Accept acidified
		1	
		[3]	
2(b)(ii)	Depends on candidate answer. Orange to green/ pink to colourless	[1]	
2(b)(iii)	Propan-2-ol/ $CH_3-CH(OH)-CH_3$	[1]	

Example candidate response—grade A

(b) It is known that L and M can be oxidised.

(i) Suggest a suitable oxidising agent and give conditions.

Reagent $K_2Cr_2O_7$ ✓✓ ($Cr_2O_7^{2-}$) [1] 15 1

Conditions H_2SO_4 (conc) + heating and reflux ✓✓ [2] 25 2

(ii) Describe the observation of the reaction between L and oxidising in b(i).

Color change from orange to green ✓✓
 $Cr_2O_7^{2-} \rightarrow Cr^{3+}$ [1] 15 1

(iii) Use the information from (a) and (b) to identify L.

compound L propan-2-ol ✓✓ [1] 15 1

Examiner's comment

The candidate correctly identifies the oxidising agent and is awarded 1 mark. They also receive 2 marks for stating the correct conditions of the reaction.

The candidate shows the correct observation for the reaction of compound L with the oxidising agent and is awarded 1 mark. They also correctly identify compound L, receiving 1 mark.

Example candidate response—grade C

(b) It is known that L and M can be oxidised.

(i) Suggest a suitable oxidising agent and give conditions.

Reagent $KMnO_4$ / $K_2Cr_2O_7$ ✓ [1] 1

Conditions acidified; heat ✓ [2] 2

(ii) Describe the observation of the reaction between L and oxidising in b(i).

ketone will be formed ✓
propanone ✓ [1] 0

(iii) Use the information from (a) and (b) to identify L.

compound L propan-2-ol ✓ [1] 1

Examiner's comment

The candidate correctly identifies the oxidising agent and is awarded 1 mark. They also receive 2 marks for stating the correct conditions of the reaction.

However, the candidate loses 1 mark for an incorrect observation of the reaction of compound L with the oxidising agent.

Example candidate response—grade E

(b) It is known that L and M can be oxidised.

M-al.

(i) Suggest a suitable oxidising agent and give conditions.

Reagent K₂MnO₇ X [1]

Conditions Heating ✓, Reflux X [2]

(ii) Describe the observation of the reaction between L and oxidising in b(i).

Colour change to green X [1]

(iii) Use the information from (a) and (b) to identify L.

compound L Propanal X [1]

For
Examiner's
Use

0

1

0

0

Examiner's comment

The candidate incorrectly identifies the oxidising agent and loses 1 mark. They receive 1 mark for stating the correct condition and lose 1 mark for the incorrect condition.

The candidate shows an incorrect observation of the reaction of compound L with the oxidising agent and lost 1 mark. They also incorrectly identify compound L, and lose 1 mark for this.

Question 2 (c), (d)

(c) Suggest the reagent which gives the same positive test with both Q and M.

..... [1]

(d) What would be observed when Q and M separately mixed with Tollen's reagent.

compound Q

compound M

[2]

Mark Scheme

Question	Answer	Mark	Additional Guidance
2(c)	2,4 -DNPH	[1]	Accept Brady's reagent and Borche's reagent
2(d)	No change with Q Silver mirror with M	1 1 [2]	

Example candidate response—grade A

(c) Suggest the reagent which gives the same positive test with both **Q** and **M**.

..... 2,4-DNPH ✓ [1]

(d) What would be observed when **Q** and **M** separately mixed with Tollen's reagent.

compound **Q** no reaction ✓

compound **M** silver mirror ✓

[2]

Examiner's comment

The candidate receives 1 mark for correctly identifying the reagent that gives a positive test with both compounds **Q** and **M**.

They correctly show the observations with both compounds **Q** and **M** and are awarded 2 marks.

Example candidate response—grade C

- (c) Suggest the reagent which gives the same positive test with both **Q** and **M**.
2,4-Diethylamide X [1] 0
- (d) What would be observed when **Q** and **M** separately mixed with Tollen's reagent.
Q → propanone
- compound **Q** No change ✓ reaction will not occur 1
- compound **M** Silver mirror ✓ occurs 1
- [2]

Examiner's comment

The candidate loses 1 mark for incorrectly identifying the reagent that gives a positive test with both compounds **Q** and **M**.

They correctly show the observations with both compounds **Q** and **M** and receive 2 marks.

Example candidate response—grade E

- (c) Suggest the reagent which gives the same positive test with both **Q** and **M**.
Brady's reagent - 2,4-dinitrophenylhydrazine (orange) ✓ [1] 1
- (d) What would be observed when **Q** and **M** separately mixed with Tollen's reagent.
- compound **Q** remain blue X
- compound **M** blue Cu²⁺ to reddish brown X
- no change [2] 0

Examiner's comment

The candidate receives 1 mark for correctly identifying the reagent that gives a positive test with both compounds **Q** and **M**.

They show incorrect observations with for both compounds **Q** and **M** and lose 1 mark.

Question 2 (e)

(e) **Q** can be reduced to **L**.

(i) Give the formula of a suitable reducing agent.

..... [1]

(ii) Write a balanced equation for the conversion of **Q** to **L** using [H] to represent the reducing agent.

..... [1]

Mark Scheme

Question	Answer	Marks	Additional Guidance
2(e)(i)	NaBH ₄ /LiAlH ₄	[1]	
2(e)(ii)	CH ₃ -CO-CH ₃ + 2[H] → CH ₃ -CH(OH)-CH ₃	[1]	

Example candidate response—grade A

(e) **Q** can be reduced to **L**.

(i) Give the formula of a suitable reducing agent.

LiAlH₄ or NaBH₄ ✓ ✓ [1]

(ii) Write a balanced equation for the conversion of **Q** to **L** using [H] to represent the reducing agent.

CH₃-C(=O)-CH₃ (aq) + 2[H] → CH₃-CH(OH)-CH₃ ✓ ✓ [1]

15 1

15 1

Examiner's comment

The candidate correctly identifies the formula for a suitable reducing agent and receives 1 mark. In addition, they provide the correct balanced equation for the reaction of compound **Q** with the reducing agent, receiving 1 mark.

Example candidate response—grade C

(e) Q can be reduced to L.

(i) Give the formula of a suitable reducing agent.

~~H₂~~ NaBH₄ ✓ [1] 1 /

(ii) Write a balanced equation for the conversion of Q to L using [H] to represent the reducing agent.

CH₃COCH₃ + [H] ⇒ CH₃CH(OH)CH₃ × [1] 0 0

Examiner's comment

The candidate correctly identifies the formula for a suitable reducing agent and receives 1 mark. However, they write an unbalanced equation for the reaction of compound Q with the reducing agent and lose 1 mark.

Example candidate response—grade E

(e) Q can be reduced to L.

(i) Give the formula of a suitable reducing agent.

H₂O × [1] 0

(ii) Write a balanced equation for the conversion of Q to L using [H] to represent the reducing agent.

CH₃COCH₃ + [H] → CH₃CH(OH)CH₃ [1] 0

Examiner's comment

The candidate incorrectly identifies the formula for a suitable reducing agent and loses 1 mark. In addition, they wrote an unbalanced equation of compound Q with the reducing agent and lose 1 mark.